

## Year 7 · Chapter 2 · Angles: answers (teacher copy)

### 2.01 Naming angles (booklet p. 3)

#### WE DO examples (teacher)

1. Column 1, E1:  $\angle PEC \rightarrow E$
2. Column 1, E2:  $\angle PAQ \rightarrow A$
3. Column 1, E3: (diagram)  $\rightarrow \angle UPY$  or  $\angle YPU$
4. Column 2, E1:  $\angle HBX \rightarrow$  also  $\angle XBH$ ; vertex B, arms BH and BX
5. Column 2, E2:  $\angle FYW \rightarrow$  also  $\angle WYF$ ; vertex Y, arms YF and YW
6. Column 3, E1: Name the three angles at E.  $\rightarrow$  small angles:  $\angle AEZ$ ,  $\angle ZEK$ ; whole angle:  $\angle AEK$
7. Column 3, E2:  $\angle PME$  and  $\angle EMY$  are next to each other. Name the vertex and the arm they share.  $\rightarrow$  vertex: M (middle letter of both); common arm: ME

#### Questions (in order: Easy, Medium, Challenging)

- |                                  |                                          |                                                 |
|----------------------------------|------------------------------------------|-------------------------------------------------|
| 1. Y                             | 15. $\angle SHU$ or $\angle UHS$         | 23. $\angle XBE$ ; vertex B; arms BE, BX        |
| 2. M                             | 16. $\angle CRZ$ ; vertex R; arms RZ, RC | 24. $\angle HAE$ ; vertex A; arms AE, AH        |
| 3. W                             | 17. $\angle QVY$ ; vertex V; arms VY, VQ | 25. $\angle WAG$ ; vertex A; arms AG, AW        |
| 4. M                             | 18. $\angle ULN$ ; vertex L; arms LN, LU | 26. $\angle NWC$ , $\angle CWL$ , $\angle NWL$  |
| 5. X                             | 19. $\angle LUW$ ; vertex U; arms UW, UL | 27. vertex M, common arm MQ                     |
| 6. X                             | 20. $\angle CLS$ ; vertex L; arms LS, LC | 28. $\angle KFM$ , $\angle MFQ$ , $\angle K FQ$ |
| 7. U                             | 21. $\angle HGV$ ; vertex G; arms GV, GH | 29. vertex U, common arm UR                     |
| 8. S                             | 22. $\angle STM$ ; vertex T; arms TM, TS |                                                 |
| 9. B                             |                                          |                                                 |
| 10. Z                            |                                          |                                                 |
| 11. H                            |                                          |                                                 |
| 12. K                            |                                          |                                                 |
| 13. $\angle GHQ$ or $\angle QHG$ |                                          |                                                 |
| 14. $\angle GPR$ or $\angle RPG$ |                                          |                                                 |

### 2.02 Measuring and drawing angles

(booklet p. 4)

#### WE DO examples (teacher)

1. Column 1, E1: (diagram)  $\rightarrow 40^\circ$
2. Column 1, E2: (diagram)  $\rightarrow 145^\circ$
3. Column 1, E3:  $285^\circ \rightarrow 75^\circ$
4. Column 2, E1: Estimate, then measure.  $\rightarrow$  estimate: about  $140^\circ$ ; measure:  $137^\circ$
5. Column 2, E2: Estimate, then measure.  $\rightarrow$  estimate: about  $40^\circ$ ; measure:  $40^\circ$
6. Column 3, E1: How do you draw a reflex angle of  $200^\circ$ ?  $\rightarrow 360^\circ - 200^\circ = 160^\circ$ ; draw  $160^\circ$ , then mark the reflex angle outside it
7. Column 3, E2: An angle is  $60^\circ$  less than a revolution. Find its size and classify it.  $\rightarrow 360^\circ - 60^\circ = 300^\circ$ ;  $300^\circ$  is reflex

#### Questions (in order: Easy, Medium, Challenging)

- |                 |                                               |                                               |
|-----------------|-----------------------------------------------|-----------------------------------------------|
| 1. $135^\circ$  | 12. $125^\circ$                               | 21. draw $50^\circ$ , mark the outside angle  |
| 2. $115^\circ$  | 13. $155^\circ$                               | 22. $285^\circ$ , reflex                      |
| 3. $80^\circ$   | 14. $85^\circ$                                | 23. draw $35^\circ$ , mark the outside angle  |
| 4. $60^\circ$   | 15. $130^\circ$                               | 24. $290^\circ$ , reflex                      |
| 5. $5^\circ$    | 16. $49^\circ$                                | 25. draw $120^\circ$ , mark the outside angle |
| 6. $35^\circ$   | 17. $121^\circ$                               |                                               |
| 7. $110^\circ$  | 18. $57^\circ$                                |                                               |
| 8. $120^\circ$  | 19. draw $135^\circ$ , mark the outside angle |                                               |
| 9. $170^\circ$  | 20. $320^\circ$ , reflex                      |                                               |
| 10. $150^\circ$ |                                               |                                               |
| 11. $160^\circ$ |                                               |                                               |

### 2.03 Classifying angles (booklet p. 5)

#### WE DO examples (teacher)

1. Column 1, E1:  $360^\circ \rightarrow$  revolution
2. Column 1, E2:  $90^\circ \rightarrow$  right
3. Column 1, E3: (diagram)  $\rightarrow$  right
4. Column 2, E1:  $155^\circ + 95^\circ \rightarrow 155^\circ + 95^\circ = 250^\circ$ ;  $250^\circ$  is reflex
5. Column 2, E2:  $150^\circ + 160^\circ \rightarrow 150^\circ + 160^\circ = 310^\circ$ ;  $310^\circ$  is reflex
6. Column 3, E1: The angle between the hands of a clock at 10:00.  $\rightarrow$  each hour mark:  $360^\circ \div 12 = 30^\circ$ ;  $2 \times 30^\circ = 60^\circ$ ; acute
7. Column 3, E2: A right angle plus  $20^\circ$ .  $\rightarrow 90^\circ + 20^\circ = 110^\circ$ ; obtuse
8. Column 3, E3: A straight angle plus  $10^\circ$ .  $\rightarrow 180^\circ + 10^\circ = 190^\circ$ ; reflex

#### Questions (in order: Easy, Medium, Challenging)

- |             |                          |                          |
|-------------|--------------------------|--------------------------|
| 1. acute    | 16. reflex               | 31. $115^\circ$ , obtuse |
| 2. straight | 17. $170^\circ$ , obtuse | 32. $175^\circ$ , obtuse |
| 3. obtuse   | 18. $330^\circ$ , reflex | 33. $215^\circ$ , reflex |
| 4. reflex   | 19. $150^\circ$ , obtuse | 34. $160^\circ$ , obtuse |
| 5. obtuse   | 20. $150^\circ$ , obtuse | 35. $130^\circ$ , obtuse |
| 6. reflex   | 21. $185^\circ$ , reflex | 36. $160^\circ$ , obtuse |
| 7. obtuse   | 22. $120^\circ$ , obtuse | 37. $140^\circ$ , obtuse |
| 8. acute    | 23. $155^\circ$ , obtuse | 38. $160^\circ$ , obtuse |
| 9. reflex   | 24. $195^\circ$ , reflex | 39. $90^\circ$ , right   |
| 10. reflex  | 25. $320^\circ$ , reflex | 40. $135^\circ$ , obtuse |
| 11. obtuse  | 26. $195^\circ$ , reflex | 41. $230^\circ$ , reflex |
| 12. reflex  | 27. $135^\circ$ , obtuse | 42. $120^\circ$ , obtuse |
| 13. right   | 28. $150^\circ$ , obtuse | 43. $100^\circ$ , obtuse |
| 14. acute   | 29. $125^\circ$ , obtuse | 44. $210^\circ$ , reflex |
| 15. reflex  | 30. $35^\circ$ , acute   | 45. $150^\circ$ , obtuse |

### 2.04 Complementary and supplementary angles (booklet p. 6)

#### WE DO examples (teacher)

1. Column 1, E1:  $75^\circ \rightarrow 15^\circ$
2. Column 1, E2:  $30^\circ \rightarrow 60^\circ$
3. Column 1, E3:  $140^\circ \rightarrow 40^\circ$
4. Column 2, E1:  $p^\circ$  and  $75^\circ$  are complementary.  $\rightarrow p + 75 = 90$ ;  $p = 90 - 75 = 15$
5. Column 2, E2:  $p^\circ$  and  $17^\circ$  are supplementary.  $\rightarrow p + 17 = 180$ ;  $p = 180 - 17 = 163$
6. Column 3, E1: Find x.  $\rightarrow 3x + 132 = 180$ ;  $3x = 48$ ;  $x = 16$
7. Column 3, E2: Find x.  $\rightarrow x + x + 10 = 90$ ;  $2x = 80$ ;  $x = 15$

#### Questions (in order: Easy, Medium, Challenging)

- |                |                 |               |
|----------------|-----------------|---------------|
| 1. $77^\circ$  | 13. $75^\circ$  | 25. $x = 45$  |
| 2. $18^\circ$  | 14. $111^\circ$ | 26. $m = 152$ |
| 3. $35^\circ$  | 15. $105^\circ$ | 27. $p = 74$  |
| 4. $20^\circ$  | 16. $44^\circ$  | 28. $m = 135$ |
| 5. $9^\circ$   | 17. $153^\circ$ | 29. $p = 54$  |
| 6. $23^\circ$  | 18. $130^\circ$ | 30. $y = 140$ |
| 7. $22^\circ$  | 19. $30^\circ$  | 31. $x = 15$  |
| 8. $65^\circ$  | 20. $106^\circ$ | 32. $p = 147$ |
| 9. $85^\circ$  | 21. $7^\circ$   | 33. $p = 50$  |
| 10. $28^\circ$ | 22. $124^\circ$ | 34. $x = 40$  |
| 11. $74^\circ$ | 23. $2^\circ$   | 35. $x = 24$  |
| 12. $5^\circ$  | 24. $161^\circ$ |               |

## 2.05 Angles at a point and vertically opposite angles (booklet p. 7)

### WE DO examples (teacher)

- Column 1, E1: (diagram)  $\rightarrow a = 120$
- Column 1, E2: (diagram)  $\rightarrow a = 45$
- Column 1, E3: (diagram)  $\rightarrow b = 80$
- Column 2, E1: Find  $a$ .  $\rightarrow a + 130 + 125 = 360$ ;  $a = 360 - 255 = 105$
- Column 2, E2: Find  $a$ .  $\rightarrow a + 65 + 105 = 360$ ;  $a = 360 - 170 = 190$
- Column 3, E1: Find  $a$ ,  $b$  and  $c$ .  $\rightarrow b = 125$  (vertically opposite);  $a = 180 - 125 = 55$  (straight line);  $c = 55$  (vertically opposite  $a$ )
- Column 3, E2: Find  $a$ ,  $b$  and  $c$ .  $\rightarrow b = 45$  (vertically opposite);  $a = 180 - 45 = 135$  (straight line);  $c = 135$  (vertically opposite  $a$ )

### Questions (in order: Easy, Medium, Challenging)

- |              |               |                                      |
|--------------|---------------|--------------------------------------|
| 1. $a = 75$  | 6. $b = 60$   | 11. $a = 65$ , $b = 115$ , $c = 65$  |
| 2. $a = 155$ | 7. $b = 70$   | 12. $a = 110$ , $b = 70$ , $c = 110$ |
| 3. $a = 110$ | 8. $a = 205$  |                                      |
| 4. $a = 115$ | 9. $a = 190$  |                                      |
| 5. $b = 140$ | 10. $a = 125$ |                                      |

## 2.06 Parallel and perpendicular lines

(booklet p. 8)

### WE DO examples (teacher)

- Column 1, E1: KL is parallel to MN  $\rightarrow KL \parallel MN$
- Column 1, E2: XY is perpendicular to ZW  $\rightarrow XY \perp ZW$
- Column 1, E3: train tracks  $\rightarrow \parallel$
- Column 2, E1: Rectangle TUVW  $\rightarrow \parallel$ : TU  $\parallel$  WV, TW  $\parallel$  UV;  $\perp$ : TU  $\perp$  UV (every corner)
- Column 2, E2: Parallelogram KLMN  $\rightarrow \parallel$ : KL  $\parallel$  NM, KN  $\parallel$  LM;  $\perp$ : none
- Column 3, E1: Construct a line perpendicular to PQ through the point G on PQ.  $\rightarrow$  compasses at G: mark two points on PQ; from each point: equal arcs that cross; rule from G through the crossing
- Column 3, E2: Construct a line parallel to PQ through the point H (not on PQ).  $\rightarrow$  rule a line through H that cuts PQ; copy the angle at H (corresponding); rule the new arm: it is  $\parallel$  to PQ
- Column 3, E3: Construct the perpendicular bisector of the interval CD.  $\rightarrow$  compasses more than half of CD; arcs from C and from D, above and below; rule through the two crossings

### Questions (in order: Easy, Medium, Challenging)

- |                      |                                                                                  |                     |
|----------------------|----------------------------------------------------------------------------------|---------------------|
| 1. EF $\parallel$ GH | 19. $\parallel$                                                                  | 26. arcs on CD      |
| 2. AB $\perp$ CD     | 20. neither                                                                      | either side of      |
| 3. PQ $\parallel$ RS | 21. $\parallel$                                                                  | W; equal arcs       |
| 4. EF $\perp$ GH     | 22. $\parallel$                                                                  | from them           |
| 5. PQ $\perp$ RS     | 23. EF $\parallel$ HG; $\perp$ : EH $\perp$ EF, EH $\perp$ HG                    | cross; rule from    |
| 6. XY $\parallel$ ZW | 24. KL $\parallel$ NM, KN $\parallel$ LM; $\perp$ : KL $\perp$ LM (every corner) | W through the       |
| 7. TU $\parallel$ VW |                                                                                  | crossing            |
| 8. TU $\perp$ VW     | 25. EF $\parallel$ HG; $\perp$ : none                                            | 27. rule a          |
| 9. AB $\parallel$ CD |                                                                                  | transversal         |
| 10. KL $\perp$ MN    |                                                                                  | through S;          |
| 11. $\perp$          |                                                                                  | copy its angle      |
| 12. neither          |                                                                                  | with PQ at S;       |
| 13. $\parallel$      |                                                                                  | rule the new        |
| 14. $\perp$          |                                                                                  | arm                 |
| 15. neither          |                                                                                  | 28. equal arcs from |
| 16. $\parallel$      |                                                                                  | E and F above       |
| 17. $\perp$          |                                                                                  | and below; rule     |
| 18. $\perp$          |                                                                                  | through the         |
|                      |                                                                                  | two crossings       |

- arcs on PQ
- either side of T;
- equal arcs from
- them cross;
- rule from T
- through the
- crossing

## 2.07 Corresponding angles (booklet p. 9)

### WE DO examples (teacher)

- Column 1, E1: corresponding to  $76^\circ \rightarrow 76^\circ$
- Column 1, E2: corresponding to  $138^\circ \rightarrow 138^\circ$
- Column 1, E3: (diagram)  $\rightarrow a = 50$
- Column 2, E1:  $a$  is corresponding to  $135^\circ \rightarrow a = 135$  (corresponding angles, parallel lines);  $b = 180 - 135 = 45$  (straight line)
- Column 2, E2:  $a$  is corresponding to  $100^\circ \rightarrow a = 100$  (corresponding angles, parallel lines);  $b = 180 - 100 = 80$  (straight line)
- Column 3, E1: Corresponding angles are  $(4x + 10)^\circ$  and  $114^\circ$ . Find  $x$ .  $\rightarrow 4x + 10 = 114$ ;  $4x = 104$ ;  $x = 26$
- Column 3, E2: Corresponding angles are  $(2x + 15)^\circ$  and  $65^\circ$ . Find  $x$ .  $\rightarrow 2x + 15 = 65$ ;  $2x = 50$ ;  $x = 25$

### Questions (in order: Easy, Medium, Challenging)

- |                |                          |                          |
|----------------|--------------------------|--------------------------|
| 1. $149^\circ$ | 11. $a = 70$             | 21. $a = 85$ , $b = 95$  |
| 2. $113^\circ$ | 12. $a = 135$            | 22. $a = 50$ , $b = 130$ |
| 3. $39^\circ$  | 13. $a = 115$ , $b = 65$ | 23. $x = 5$              |
| 4. $51^\circ$  | 14. $a = 145$ , $b = 35$ | 24. $x = 15$             |
| 5. $28^\circ$  | 15. $a = 110$ , $b = 70$ | 25. $x = 8$              |
| 6. $43^\circ$  | 16. $a = 80$ , $b = 100$ | 26. $x = 16$             |
| 7. $118^\circ$ | 17. $a = 95$ , $b = 85$  | 27. $x = 28$             |
| 8. $109^\circ$ | 18. $a = 125$ , $b = 55$ | 28. $x = 30$             |
| 9. $65^\circ$  | 19. $a = 150$ , $b = 30$ |                          |
| 10. $a = 50$   | 20. $a = 70$ , $b = 110$ |                          |

## 2.08 Alternate angles (booklet p. 10)

### WE DO examples (teacher)

- Column 1, E1: alternate to  $65^\circ \rightarrow 65^\circ$
- Column 1, E2: alternate to  $33^\circ \rightarrow 33^\circ$
- Column 1, E3: (diagram)  $\rightarrow a = 60$
- Column 2, E1:  $a$  is alternate to  $120^\circ \rightarrow a = 120$  (alternate angles, parallel lines);  $b = 180 - 120 = 60$  (straight line)
- Column 2, E2:  $a$  is alternate to  $70^\circ \rightarrow a = 70$  (alternate angles, parallel lines);  $b = 180 - 70 = 110$  (straight line)
- Column 3, E1: Alternate angles are  $(4x - 15)^\circ$  and  $61^\circ$ . Find  $x$ .  $\rightarrow 4x - 15 = 61$ ;  $4x = 76$ ;  $x = 19$
- Column 3, E2: Alternate angles are  $(3x - 25)^\circ$  and  $74^\circ$ . Find  $x$ .  $\rightarrow 3x - 25 = 74$ ;  $3x = 99$ ;  $x = 33$

### Questions (in order: Easy, Medium, Challenging)

- |                 |                          |                          |
|-----------------|--------------------------|--------------------------|
| 1. $127^\circ$  | 12. $36^\circ$           | 23. $a = 40$ , $b = 140$ |
| 2. $108^\circ$  | 13. $a = 110$            | 24. $a = 45$ , $b = 135$ |
| 3. $82^\circ$   | 14. $a = 50$             | 25. $a = 30$ , $b = 150$ |
| 4. $159^\circ$  | 15. $a = 25$             | 26. $x = 9$              |
| 5. $154^\circ$  | 16. $a = 155$ , $b = 25$ | 27. $x = 26$             |
| 6. $48^\circ$   | 17. $a = 80$ , $b = 100$ | 28. $x = 18$             |
| 7. $66^\circ$   | 18. $a = 105$ , $b = 75$ | 29. $x = 32$             |
| 8. $24^\circ$   | 19. $a = 100$ , $b = 80$ | 30. $x = 30$             |
| 9. $133^\circ$  | 20. $a = 140$ , $b = 40$ | 31. $x = 17$             |
| 10. $156^\circ$ | 21. $a = 55$ , $b = 125$ |                          |
| 11. $106^\circ$ | 22. $a = 125$ , $b = 55$ |                          |

## 2.09 Co-interior angles (booklet p. 11)

### WE DO examples (teacher)

- Column 1, E1: co-interior to  $59^\circ \rightarrow 121^\circ$
- Column 1, E2: co-interior to  $128^\circ \rightarrow 52^\circ$

3. Column 1, E3: (diagram)  $\rightarrow a = 45$
4. Column 2, E1:  $a$  is co-interior to  $38^\circ$ .  $\rightarrow a + 38 = 180$  (co-interior angles);  $a = 180 - 38 = 142$
5. Column 2, E2:  $a$  is co-interior to  $94^\circ$ .  $\rightarrow a + 94 = 180$  (co-interior angles);  $a = 180 - 94 = 86$
6. Column 3, E1: Co-interior angles are  $y^\circ$  and  $2y^\circ$ . Find  $y$  and both angles.  $\rightarrow y + 2y = 180$ ;  $3y = 180$ , so  $y = 60$ ; angles:  $60^\circ$  and  $120^\circ$
7. Column 3, E2: Co-interior angles are  $y^\circ$  and  $3y^\circ$ . Find  $y$  and both angles.  $\rightarrow y + 3y = 180$ ;  $4y = 180$ , so  $y = 45$ ; angles:  $45^\circ$  and  $135^\circ$

#### Questions (in order: Easy, Medium, Challenging)

- |                |                |                                           |
|----------------|----------------|-------------------------------------------|
| 1. $23^\circ$  | 10. $40^\circ$ | 19. $a = 150$                             |
| 2. $36^\circ$  | 11. $77^\circ$ | 20. $a = 153$                             |
| 3. $134^\circ$ | 12. $61^\circ$ | 21. $a = 103$                             |
| 4. $125^\circ$ | 13. $a = 70$   | 22. $a = 44$                              |
| 5. $38^\circ$  | 14. $a = 140$  | 23. $a = 63$                              |
| 6. $139^\circ$ | 15. $a = 115$  | 24. $a = 135$                             |
| 7. $149^\circ$ | 16. $a = 111$  | 25. $a = 109$                             |
| 8. $48^\circ$  | 17. $a = 123$  | 26. $y = 36$ ; $36^\circ$ and $144^\circ$ |
| 9. $66^\circ$  | 18. $a = 77$   |                                           |

## 2.10 Finding unknown angles on parallel lines (booklet p. 12)

#### WE DO examples (teacher)

1. Column 1, E1: F shape  $\rightarrow$  corresponding
2. Column 1, E2: Z shape  $\rightarrow$  alternate
3. Column 1, E3: (diagram)  $\rightarrow a = 45$
4. Column 2, E1: Find  $a$ .  $\rightarrow a = 70$ ; corresponding angles are equal
5. Column 2, E2: Find  $a$ .  $\rightarrow a = 140$ ; alternate angles are equal
6. Column 2, E3: Find  $a$ .  $\rightarrow a = 180 - 140 = 40$ ; co-interior angles add to  $180^\circ$
7. Column 3, E1: Find  $y$ , then  $x$ .  $\rightarrow y = 180 - 45 = 135$  (straight line);  $x = 135$  (corresponding to  $y$ )

8. Column 3, E2: Find  $y$ , then  $x$ .  $\rightarrow y = 180 - 95 = 85$  (straight line);  $x = 85$  (corresponding to  $y$ )

#### Questions (in order: Easy, Medium, Challenging)

- |                  |                  |                           |
|------------------|------------------|---------------------------|
| 1. co-interior   | 7. corresponding | 12. $a = 45$              |
| 2. corresponding | 8. $a = 35$      | (corresponding)           |
| 3. alternate     | 9. $a = 120$     | 13. $a = 50$              |
| 4. co-interior   | 10. $a = 75$     | (alternate)               |
| 5. co-interior   | 11. $a = 150$    | 14. $y = 90$ , $x = 90$   |
| 6. alternate     |                  | 15. $y = 95$ , $x = 95$   |
|                  |                  | 16. $y = 130$ , $x = 130$ |

## 2.11 Proving lines are parallel (booklet p. 13)

#### WE DO examples (teacher)

1. Column 1, E1: alternate:  $75^\circ$  and  $105^\circ \rightarrow$  no
2. Column 1, E2: co-interior:  $80^\circ$  and  $100^\circ \rightarrow$  yes
3. Column 2, E1: Corresponding angles:  $60^\circ$  and  $65^\circ \rightarrow 60 \neq 65$ ; no: they are not equal
4. Column 2, E2: Alternate angles:  $75^\circ$  and  $80^\circ \rightarrow 75 \neq 80$ ; no: they are not equal
5. Column 2, E3: Co-interior angles:  $130^\circ$  and  $50^\circ \rightarrow 130 + 50 = 180$ ; yes: co-interior angles add to  $180^\circ$
6. Column 3, E1: Alternate angles are  $(3x + 10)^\circ$  and  $97^\circ$ .  $\rightarrow 3x + 10 = 97$ ;  $3x = 87$ ;  $x = 29$
7. Column 3, E2: Co-interior angles are  $2x^\circ$  and  $160^\circ$ .  $\rightarrow 2x + 160 = 180$ ;  $2x = 20$ ;  $x = 10$

#### Questions (in order: Easy, Medium, Challenging)

- |        |         |              |
|--------|---------|--------------|
| 1. yes | 12. yes | 23. yes      |
| 2. no  | 13. no  | 24. yes      |
| 3. yes | 14. yes | 25. no       |
| 4. yes | 15. yes | 26. yes      |
| 5. yes | 16. yes | 27. $x = 17$ |
| 6. no  | 17. no  | 28. $x = 15$ |
| 7. yes | 18. yes | 29. $x = 6$  |
| 8. no  | 19. yes | 30. $x = 17$ |
| 9. yes | 20. yes | 31. $x = 8$  |
| 10. no | 21. yes | 32. $x = 12$ |
| 11. no | 22. yes |              |